E-COMMERCE SALES & CUSTOMER ANALYTICS

BUSINESS ANALYSIS REPORT

A Full-Cycle Data Analytics Case Study

GROUP 2

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1. Introduction

This report presents an end-to-end data analysis of an e-commerce company's dataset with customer, product, order, and location data. As part of an internship case study, the analysis was carried out as a Junior Data Analyst would approach a real business request: identifying data-quality issues, cleaning the raw datasets, and deriving evidence-based insights using a combination of tools — Excel/Power Query, Power BI, MySQL, and Python (Pandas, Matplotlib, Seaborn) — reflecting the range of tools typically used in a professional analytics workflow rather than relying on a single platform.

The underlying dataset consists of four related files: customers (4,010 raw records), orders (18,020 raw records), products (30 records), and locations (8 records), covering transactions across eight Indian cities over the 2025 calendar year. After cleaning, the working dataset contains 4,000 customers and 18,000 orders.

The analysis uses a consistent all-order basis across the report, with cancellation and return performance analysed separately as an important operational KPI.

2. Problem Statement

The company's management needs a clear understanding of sales performance, customer behaviour, product performance, and the impact of discounting on sales, but the underlying data — spread across four separate files — had not been validated or prepared for analysis. Raw data quality could not be assumed: it needed to be independently inspected, cleaned, and documented before any business conclusions could be drawn.

The core problem this project addresses is: what does the data actually reveal about sales, customers, products, and discounts, once it has been properly cleaned and correctly analyzed — and what should management do differently as a result?

3. Objectives

- Assess data quality across all four datasets and identify missing values, duplicates, invalid entries, and formatting inconsistencies.
- Clean and standardize the data, documenting every cleaning decision and its justification.
- Answer a defined set of business questions covering sales trends, product and category performance, customer segmentation, location performance, and discount impact.
- Build relational data models and interactive dashboards (Power BI, Excel) to communicate findings visually.
- Validate findings using SQL and Python analysis, cross-checking results across tools for consistency.
- Distinguish clearly between facts established by the data and recommendations proposed by the team.

4. Methodology

The project followed a five-phase workflow, moving from raw data to a fully documented, multi-tool analysis:

Phase A — Data Cleaning (Excel/Power Query): Raw CSV files were loaded into Power Query, inspected for missing values, duplicates, and inconsistent formatting, and cleaned with fully documented, repeatable steps.

Phase B — Business Questions (Excel PivotTables & Charts): Eight core business questions were answered using PivotTables and PivotCharts built on the cleaned data model.

Phase C — SQL Analysis (MySQL): The cleaned data was imported into a relational MySQL database with proper foreign keys, and seven further business questions were answered using joins, aggregation, and filtering.

Phase D — Python Analysis (Pandas, NumPy, Matplotlib, Seaborn): An independent cleaning and analysis pass was carried out in Python, including descriptive statistics, outlier detection, groupby analysis, correlation analysis, and the required visualizations.

Phase E — Power BI Dashboard: An interactive four-page Power BI report was built on a star-schema data model with core DAX measures and additional date-intelligence measures.

5. Key Definitions & Methodology Decisions

To maintain consistency across all analytical outputs, Total Sales is defined using the complete set of 18,000 cleaned orders. This section documents the final decisions made so that every number elsewhere in this report is built on one consistent definition.

5.1 Total Sales

Decision: Total Sales = sum of Net_Sales across all 18,000 orders in the cleaned transaction dataset.

This all-order basis is used consistently across the Excel, SQL, Python and Power BI analysis. Cancellation and returned orders are retained in the transaction-level sales analysis, while their frequency is separately evaluated through the cancellation/return rate KPI.

Analysis Basis	Total Sales	Orders	Average Order Value
All orders	₹ 10,04,51,357.55	18,000 Nos	₹ 5,580.63

5.2 Discount Definition

Discount percentage is calculated for each order as ((Discount/Gross_Amount) × 100). The discount analysis compares order volume, sales and Average Order Value across the observed discount levels in the dataset. Higher discount levels are evaluated against order value and realized sales to understand the relationship between discount depth and transaction performance.

5.3 Unit_Price Correction

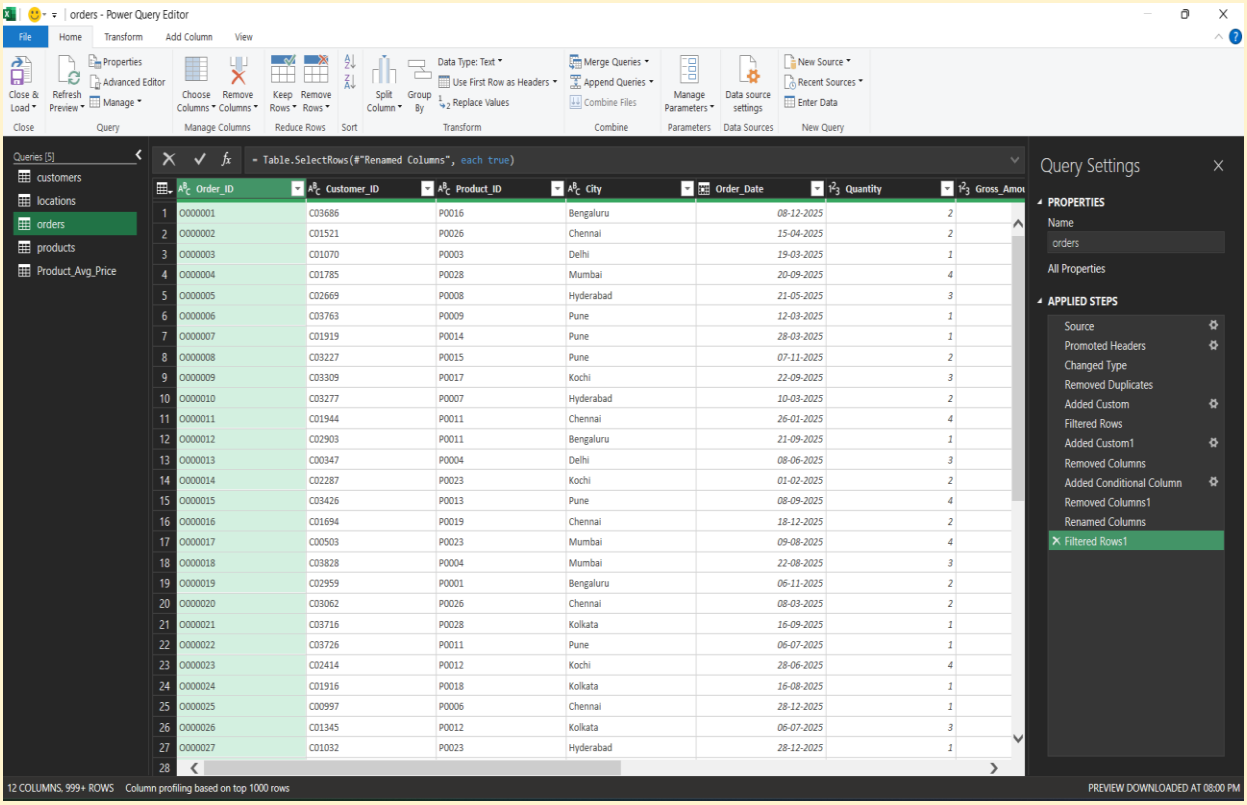
Five products (Wireless Earbuds, Bluetooth Speaker, Jeans, Handbag, Storage Box) had an invalid placeholder price of (-50) in the raw data. The correct method recovers the true price from transaction history (Gross_Amount/Quantity, which is constant across every order for a given product, confirming it is the real price, not an estimate), rather than simply taking the absolute value of (-50), which would incorrectly set the price to ₹ 50.

Product	Raw value	Correct recovered price
Wireless Earbuds	-50	₹3,621
Bluetooth Speaker	-50	₹3,441
Jeans	-50	₹4,727
Handbag	-50	₹2,559
Storage Box	-50	₹929

6. Phase A - Data Cleaning Summary

The table below summarizes every data quality issue identified in the raw data and the method used to resolve it.

Dataset	Issue Found	Resolution
customers	10 duplicate rows (4,010 → 4,000)	Removed using Power Query Remove Duplicates
customers	"Regular" / "regular" case inconsistency in Customer_Segment	Standardized to Title Case (3 clean values: New, Regular, Premium)
orders	20 duplicate rows (18,020 → 18,000)	Removed using Power Query Remove Duplicates
orders	25 blank Net_Sales values	Recomputed using Net_Sales = Gross_Amount − Discount + Shipping_Fee (formula held exactly for all non-blank rows)
orders	"Completed" / "completed" case inconsistency in Order_Status	Standardized to Title Case (3 clean values: Completed, Cancelled, Returned)
orders	Order_Date stored as text	Converted to proper Date type
products	5 products with invalid Unit_Price = (-50)	Recovered true price from Gross_Amount/Quantity (see Section 5.3) — not simply the absolute value
locations	None found	No cleaning required



Power Query Applied Steps pane, showing the full cleaning step sequence for the Orders and Customers queries

7. Phase B — Business Questions (Excel / Power Query)

The following questions were answered using PivotTables and charts in Excel, built on the cleaned data model. All sales figures below use the all-order basis defined in Section 5.1.

Q1. What is the total sales and average order value?

Total Sales: ₹ 10,04,51,357.55 across 18,000 orders. Average Order Value: ₹ 5,580.63.

Question 1 — What is the total sales and average order value?

TOTAL SALES	AVERAGE ORDER VALUE
100451357.6	5580.630975

KEY INSIGHT

The total sales were ₹100.45 million, with an average order value of ₹5,580.63 across 18,000 orders.

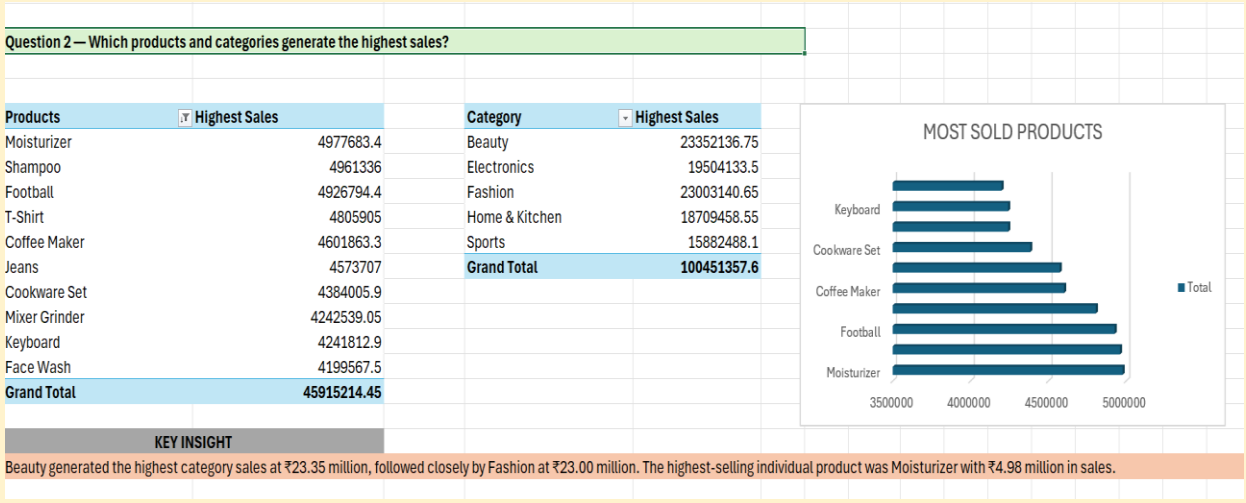
Excel PivotTable — Total Sales and Average Order Value summary

Q2. Which products and categories generate the highest sales?

Beauty (₹2.34 Cr) and Fashion (₹2.30 Cr) are the leading categories by sales, followed by Electronics (₹1.95 Cr), Home & Kitchen (₹1.87 Cr), and Sports (₹1.59 Cr).

The top individual products by sales are Moisturizer (₹49.78L), Shampoo (₹49.61L), Football (₹49.27L), T-Shirt (₹48.06L), and Coffee Maker (₹46.02L), followed by Jeans, Cookware Set, Mixer Grinder, Keyboard, and Face Wash.

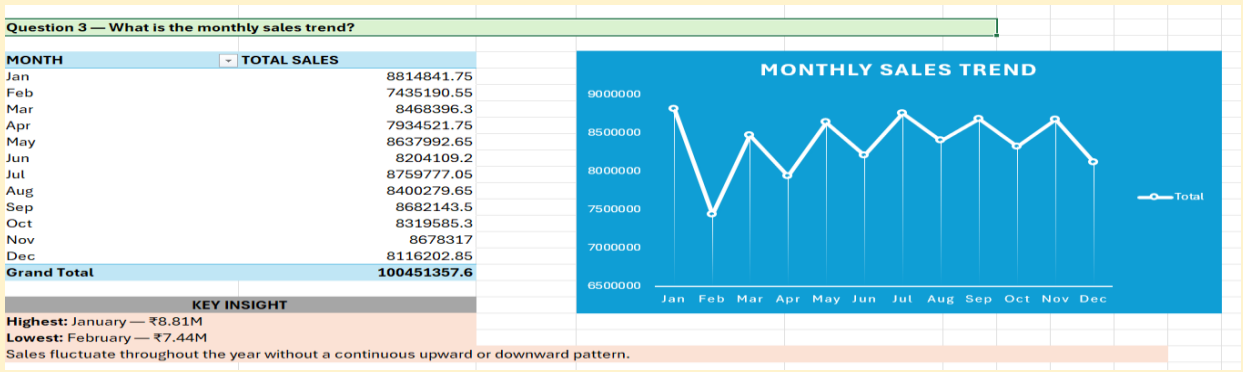
Revenue differences between products are associated mainly with unit price rather than strong demand concentration, since quantity sold is relatively uniform across products.



Excel bar charts — Sales by Category and Top 10 Products by Sales

Q3. What is the monthly sales trend?

Sales remained relatively stable throughout 2025, with moderate month-to-month variation rather than a sustained upward or downward trend. January recorded the highest monthly sales at ₹88.15 lakh, while February recorded the lowest at ₹74.35 lakh. The difference between the peak and lowest month was approximately ₹13.80 lakh.



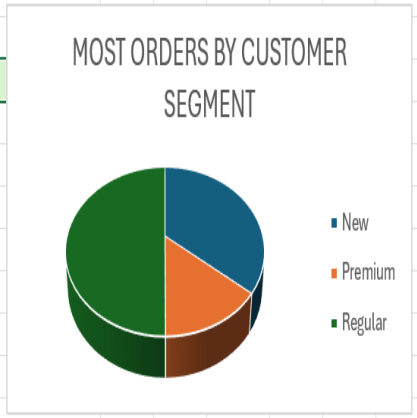
Excel line chart — Monthly Sales Trend

Q4. Which customer segments place the most orders?

Regular customers place the most orders (9,013), followed by New customers (6,424) and Premium customers (2,563). The distribution reflects the larger number of customers represented in the Regular segment.

Question 4 — Which customer segments place the most orders?

CUSTOMER SEGMENT	MOST ORDERS
New	6424
Premium	2563
Regular	9013
Grand Total	18000



KEY INSIGHT

The **Regular customer segment** placed the most orders, with **9,013 orders**, followed by New customers with 6,424 orders and Premium customers with 2,563 orders.

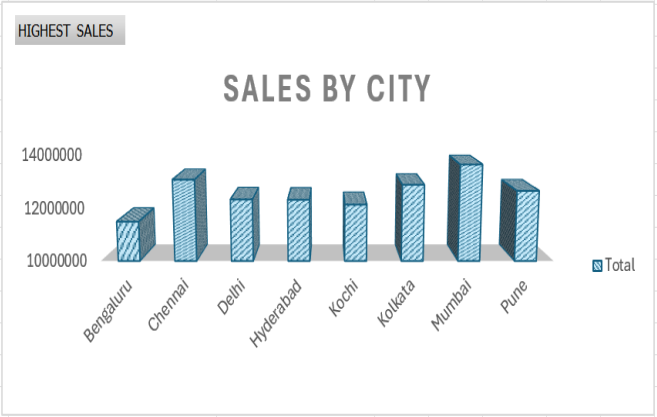
Excel column chart — Orders by Customer Segment

Q5. Which locations generate the highest sales?

Mumbai generates the highest sales at approximately ₹1.36 Cr, followed by Chennai (₹1.31 Cr), Kolkata (₹1.29 Cr), Pune (₹1.26 Cr), Delhi (₹1.23 Cr), Hyderabad (₹1.23 Cr), Kochi (₹1.21 Cr), and Bengaluru (₹1.15 Cr). Sales are relatively well distributed across the eight cities, with no single city overwhelmingly dominating the market.

Question 5 — Which locations generate the highest sales?

CITY	HIGHEST SALES
Bengaluru	11493705.3
Chennai	13059700.1
Delhi	12324678.9
Hyderabad	12311928.4
Kochi	12132312.2
Kolkata	12871309.55
Mumbai	13614514.45
Pune	12643208.65
Grand Total	100451357.6



KEY INSIGHT

Mumbai generated the highest sales, with approximately **₹13.61 million**, followed by Chennai and Kolkata.

Excel bar chart — Sales by City and Region

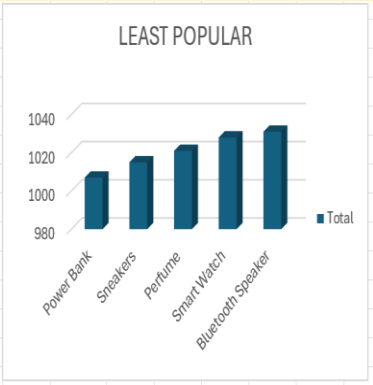
Q6. Which products are the most and least popular?

Quantity sold is nearly uniform across all 30 products, ranging only from 1,007 to 1,204 units each (about a 20% spread end-to-end, but a much smaller spread within the bulk of products). Product-level quantity sold is relatively evenly distributed, with no product accounting for a disproportionately large share of unit volume. This suggests that sales differences are influenced more by unit price than by extreme differences in product demand.

Top by quantity: Face Wash (1,204), Moisturizer (1,181), Running Shoes and Cricket Bat (1,136 each), Storage Box (1,128). Bottom by quantity: Power Bank (1,007), Sneakers (1,015), Perfume (1,021), Smart Watch (1,028), Bluetooth Speaker (1,031).

Question 6 — Which products are the most and least popular?

LEAST POPULAR		MOST POPULAR	
PRODUCT NAME	QUANTITY SOLD	PRODUCT NAME	QUANTITY SOLD
Power Bank	1007	Storage Box	1128
Sneakers	1015	Cricket Bat	1136
Perfume	1021	Running Shoes	1136
Smart Watch	1028	Moisturizer	1181
Bluetooth Speaker	1031	Face Wash	1204
Grand Total	5102	Grand Total	5785



KEY INSIGHT

Based on total quantity sold, Face Wash was the most popular product with 1,204 units sold, while Power Bank was the least popular with 1,007 units sold.

Excel PivotTable — Top 10 / Bottom 10 Products by Quantity

Q7. How do discounts affect sales and order value?

Average Order Value declines as discount depth increases. AOV falls from approximately ₹5,972 at 0% discount to ₹4,462 at 30% discount, with lower AOV observed at the intermediate discount levels as well.

Average Gross Amount increases slightly across the discount levels, but the increase is not sufficient to offset the larger discount amount. As a result, higher discount levels are associated with lower realized Net Sales per order. This is an observed relationship in the dataset rather than evidence of causation.

Question 7 — How do discounts affect sales and order value?

Discount	Orders	Total sales	Average Order Value	Total Discount
0%	6375	38069785	5971.73098	0
10%	3960	21672986.5	5472.976389	2383518.5
15%	2221	11749291.3	5290.090635	2051246.7
20%	1913	9503878.1	4968.04919	2166516.9
30%	882	3935775.4	4462.330385	1665126.6
5%	2649	15519641.3	5858.679219	808993.75
Grand Total	18000	100451358	5580.630975	9075402.45

KEY INSIGHT

Higher discount levels are associated with lower average order values in this dataset. Orders with no discount had an average order value of ₹5,971.73, compared with ₹4,462.33 for orders with a 30% discount. This indicates that larger discounts are associated with lower net order values, although the analysis does not establish that discounts themselves caused the decrease.

Excel column chart — Average Order Value by Discount

Q8. What percentage of orders are cancelled or returned?

12.24% of all orders were cancelled or returned overall (Completed: 15,797; Cancelled: 1,104; Returned: 1,099, out of 18,000 total orders).

Question 8 — What percentage of orders are cancelled or returned?

Sum of Cancelled_Returned	Orders	Percentage %
2203	18000	12.24%

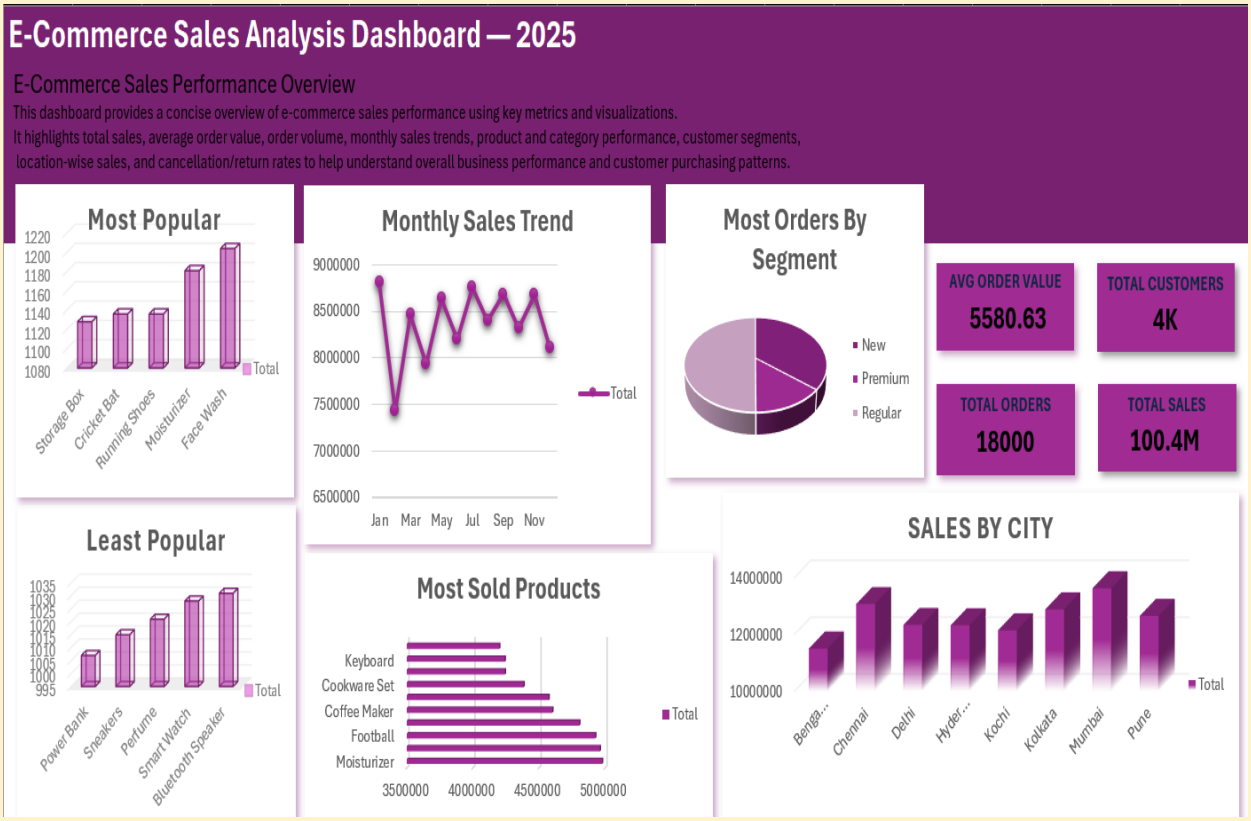
KEY INSIGHT

12.24% of all orders were either cancelled or returned, representing 2,203 out of 18,000 orders.

Excel donut chart — Order Status breakdown

Excel Dashboard Summary Page

A consolidated one-page dashboard was built summarizing the KPI cards and the four required charts for quick reference.



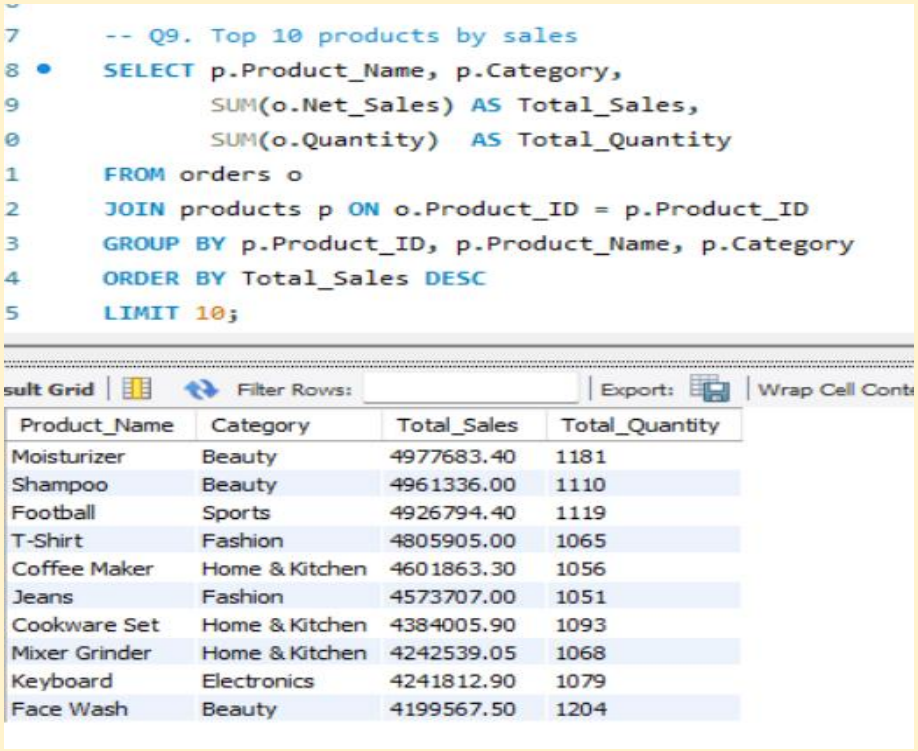
Excel Dashboard sheet — full-page summary view

8. Phase C — SQL Analysis (MySQL)

The cleaned data was imported into a MySQL database (ecommerce_analytics) with a relational structure consisting of customers, products, locations, and orders. The SQL analysis uses the complete set of 18,000 cleaned orders and applies aggregation, joins, grouping, and filtering to answer the required business questions.

Q9. Find the top 10 products by sales.

The top-selling products are Moisturizer (₹49.78L), Shampoo (₹49.61L), Football (₹49.27L), T-Shirt (₹48.06L), and Coffee Maker (₹46.02L). The remaining products in the top 10 are Jeans, Cookware Set, Mixer Grinder, Keyboard, and Face Wash. The leading products span multiple categories, indicating that sales are not concentrated in a single product category.



```
-- Q9. Top 10 products by sales
SELECT p.Product_Name, p.Category,
       SUM(o.Net_Sales) AS Total_Sales,
       SUM(o.Quantity)  AS Total_Quantity
FROM orders o
JOIN products p ON o.Product_ID = p.Product_ID
GROUP BY p.Product_ID, p.Product_Name, p.Category
ORDER BY Total_Sales DESC
LIMIT 10;
```

sult Grid | Filter Rows: | Export: | Wrap Cell Cont

Product_Name	Category	Total_Sales	Total_Quantity
Moisturizer	Beauty	4977683.40	1181
Shampoo	Beauty	4961336.00	1110
Football	Sports	4926794.40	1119
T-Shirt	Fashion	4805905.00	1065
Coffee Maker	Home & Kitchen	4601863.30	1056
Jeans	Fashion	4573707.00	1051
Cookware Set	Home & Kitchen	4384005.90	1093
Mixer Grinder	Home & Kitchen	4242539.05	1068
Keyboard	Electronics	4241812.90	1079
Face Wash	Beauty	4199567.50	1204

MySQL Workbench — Query 1 result grid (Top 10 Products by Sales)

Q10. Find the top 10 customers by purchase value.

The highest-value customer is C01258 (New segment, Mumbai), with total spending of ₹1,06,381.50 across 18 orders. This is followed by C00620 (New, Mumbai) at ₹1,01,709.35 across 13 orders and C02617 (Regular, Chennai) at ₹90,374.70 across 10 orders. C02034 (Premium, Kolkata) ranks fourth with ₹85,436.40 across 12 orders.

The top customers are distributed across New, Regular, and Premium segments, indicating that high individual customer value is not restricted to the Premium segment.

```sql
-- Q10. Top 10 customers by purchase value
SELECT c.Customer_ID, c.Customer_Segment, c.City,
       SUM(o.Net_Sales) AS Total_Spend,
       COUNT(*)         AS Total_Orders
FROM orders o
JOIN customers c ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Segment, c.City
ORDER BY Total_Spend DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Segment	City	Total_Spend	Total_Orders
C01258	New	Mumbai	106381.50	18
C00620	New	Mumbai	101709.35	13
C02617	Regular	Chennai	90374.70	10
C02034	Premium	Kolkata	85436.40	12
C00622	Regular	Mumbai	83823.35	8
C01788	Premium	Kolkata	83605.25	11
C03949	New	Bengaluru	81634.95	11
C01655	New	Chennai	81570.20	11
C01164	Regular	Hyderabad	76726.55	8
C03266	Regular	Kochi	76706.00	12

MySQL Workbench — Query 2 result grid (Top 10 Customers by Purchase Value)

Q11. Calculate sales by product category and location.

Sales are distributed across both cities and product categories. Beauty and Fashion generally contribute the strongest sales across the locations shown, while Sports records comparatively lower sales in several cities. The city-category breakdown provides a useful basis for identifying local category strengths and opportunities for targeted assortment and promotional strategies.

```sql
-- Q11. Sales by product category and location
SELECT l.City, p.Category,
       SUM(o.Net_Sales) AS Total_Sales,
       COUNT(*)         AS Orders
FROM orders o
JOIN products p  ON o.Product_ID = p.Product_ID
JOIN locations l ON o.City = l.City
GROUP BY l.City, p.Category
ORDER BY l.City, Total_Sales DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

City	Category	Total_Sales	Orders
Bengaluru	Beauty	2713858.20	438
Bengaluru	Fashion	2550081.60	421
Bengaluru	Electronics	2312525.45	418
Bengaluru	Home & Kitchen	2177570.85	455
Bengaluru	Sports	1739669.20	421
Chennai	Beauty	3050236.35	470
Chennai	Fashion	3000095.55	445
Chennai	Electronics	2576625.50	458
Chennai	Home & Kitchen	2427748.35	453
Chennai	Sports	2004994.35	455
Delhi	Beauty	3078890.90	465
Delhi	Fashion	2723688.80	402
Delhi	Electronics	2365646.15	432
Delhi	Home & Kitchen	2254199.80	429
Delhi	Sports	1902253.25	443
Hyderabad	Beauty	2935637.85	464

MySQL Workbench — Query 3 result grid (Sales by Category and Location)

Q12. Calculate monthly sales.

Monthly sales remain relatively stable throughout 2025, with January recording the highest sales at ₹88.15 lakh and February the lowest at ₹74.35 lakh. The remaining months fluctuate within a relatively narrow range, indicating no strong sustained seasonal trend.

```
237     -- Q12. Monthly sales
238     SELECT DATE_FORMAT(Order_Date, '%Y-%m') AS Order_Month,
239            SUM(Net_Sales) AS Total_Sales,
240            COUNT(*)       AS Orders
241     FROM orders
242     GROUP BY Order_Month
243     ORDER BY Order_Month;
244
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_Month	Total_Sales	Orders
2025-01	8814841.75	1585
2025-02	7435190.55	1348
2025-03	8468396.30	1471
2025-04	7934521.75	1428
2025-05	8637992.65	1537
2025-06	8204109.20	1496
2025-07	8759777.05	1536
2025-08	8400279.65	1525
2025-09	8682143.50	1579
2025-10	8319585.30	1504
2025-11	8678317.00	1520
2025-12	8116202.85	1471

MySQL Workbench — Query 4 result grid (Monthly Sales)

Q13. Find customers with more than 20 orders.

No customer exceeded 20 orders during the analysis period. The highest observed order count was 18, indicating that there was no customer with exceptionally high order frequency relative to the defined threshold.

```
44
45      -- Q13. Customers with more than 20 orders
46      -- NOTE: in this dataset, no customer places more than 18 orders --
47      -- this query correctly returns 0 rows. That is itself a finding
48      -- (no customer shows very heavy repeat-purchase behaviour here).
49      SELECT c.Customer_ID, c.Customer_Segment,
50             COUNT(*) AS Order_Count,
51             SUM(o.Net_Sales) AS Total_Spend
52      FROM orders o
53      JOIN customers c ON o.Customer_ID = c.Customer_ID
54      GROUP BY c.Customer_ID, c.Customer_Segment
55      HAVING COUNT(*) > 20
56      ORDER BY Order_Count DESC;
57
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Segment	Order_Count	Total_Spend

MySQL Workbench — Query 5 result (empty set) and the supporting Top-10-by-order-count query

Q14. Calculate the cancellation/return rate by product category.

Sports has the highest cancellation/return rate at 13.69%, followed by Electronics at 12.34%, Fashion at 12.18%, Home & Kitchen at 11.93%, and Beauty at 11.07%. Sports therefore represents the category with the highest observed cancellation/return rate.

```
-- Q14. Cancellation/return rate by product category
SELECT p.Category,
       COUNT(*) AS Total_Orders,
       SUM(CASE WHEN o.Order_Status IN ('Cancelled','Returned') THEN 1 ELSE 0 END) AS Cancelled_Or_Returned,
       ROUND(
         SUM(CASE WHEN o.Order_Status IN ('Cancelled','Returned') THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
       2) AS Cancel_Return_Rate_Pct
FROM orders o
JOIN products p ON o.Product_ID = p.Product_ID
GROUP BY p.Category
ORDER BY Cancel_Return_Rate_Pct DESC;
```

Filter Rows: | Export: | Wrap Cell Content:

Category	Total_Orders	Cancelled_Or_Returned	Cancel_Return_Rate_Pct
Sports	3667	502	13.69
Electronics	3510	433	12.34
Fashion	3473	423	12.18
Home & Kitchen	3655	436	11.93
Beauty	3695	409	11.07

MySQL Workbench — Query 6 result grid (Cancellation/Return Rate by Category)

Q15. Compare sales across different customer segments.

Regular customers generate the highest total sales at ₹5.06 Cr across 9,013 orders, followed by New customers at ₹3.57 Cr across 6,424 orders and Premium customers at ₹1.42 Cr across 2,563 orders. Average Order Value is relatively similar across the three segments: ₹5,611 for Regular, ₹5,554 for New, and ₹5,539 for Premium. This indicates that the higher total sales of the Regular segment are primarily associated with its larger order volume rather than substantially higher order value.

```
-- Q15. Compare sales across different customer segments
SELECT c.Customer_Segment,
       SUM(o.Net_Sales)                AS Total_Sales,
       COUNT(*)                        AS Total_Orders,
       ROUND(SUM(o.Net_Sales)/COUNT(*), 2) AS Avg_Order_Value,
       COUNT(DISTINCT c.Customer_ID)   AS Distinct_Customers
FROM orders o
JOIN customers c ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_Segment
ORDER BY Total_Sales DESC;
```

Filter Rows: | Export: | Wrap Cell Content:

Customer_Segment	Total_Sales	Total_Orders	Avg_Order_Value	Distinct_Customers
Regular	50573637.30	9013	5611.19	1987
New	35681577.20	6424	5554.42	1421
Premium	14196143.05	2563	5538.88	554

MySQL Workbench — Query 7 result grid (Sales Comparison by Customer Segment)

SQL Bonus: Overall KPIs

The overall SQL KPI cross-check confirms total sales of ₹10,04,51,357.55 across 18,000 orders, producing an Average Order Value of ₹5,580.63. The customer master contains 4,000 unique customers. These figures provide the overall baseline used across the analytical deliverables.

```
-- ------------------------------------------------------------
-- Bonus: overall KPIs (cross-check against Excel dashboard / Python notebook)
-- ------------------------------------------------------------
SELECT
    SUM(Net_Sales)                          AS Total_Sales,
    COUNT(*)                                AS Total_Orders,
    ROUND(SUM(Net_Sales)/COUNT(*), 2)       AS Avg_Order_Value,
    (SELECT COUNT(*) FROM customers)        AS Total_Customers
FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Sales	Total_Orders	Avg_Order_Value	Total_Customers
100451357.55	18000	5580.63	4000

MySQL Workbench — Bonus KPI result grid (Total Sales, Total Orders, Avg Order Value, Total Customers)

9. Phase D — Python Analysis Summary

9.1 Descriptive Statistics (All Orders)

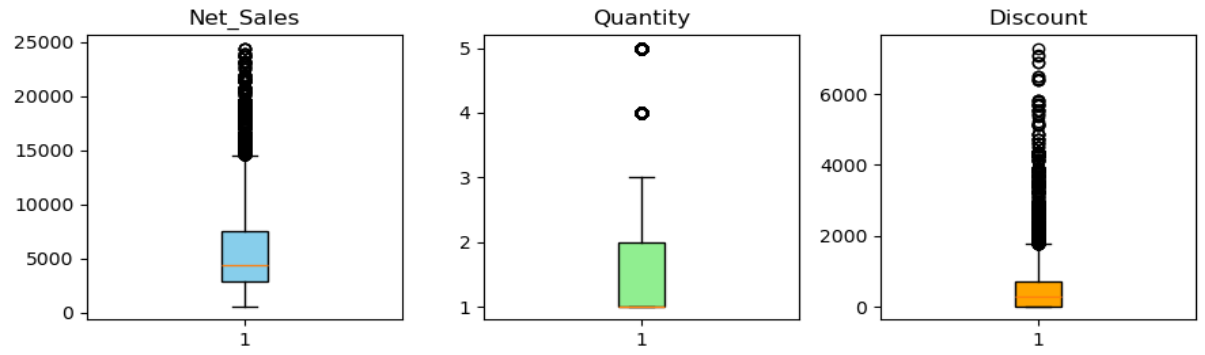
Metric	Net_Sales (₹)	Quantity	Discount (₹)	Gross_Amount (₹)
Mean	5,580.63	1.80	504.19	6,028.78
Median	4,343.00	1.00	283.20	4,722.00
Min	554.40	1	0.00	792.00
Max	24,420.00	5	7,296.00	24,320.00

Mean Net_Sales is noticeably higher than the median, indicating a right-skewed distribution: most orders are modest in value, with a smaller number of high-value orders pulling the average up.

9.2 Outlier Detection (IQR method)

- Net_Sales: approximately 630 outliers (orders above ₹14,554)
- Quantity: approximately 1,379 outliers (orders of 4–5 units)
- Discount: approximately 993 outliers (discounts above ₹1,773)

These were retained as genuine high-value or high-quantity transactions rather than treated as data errors, since they reflect real purchasing behaviour.

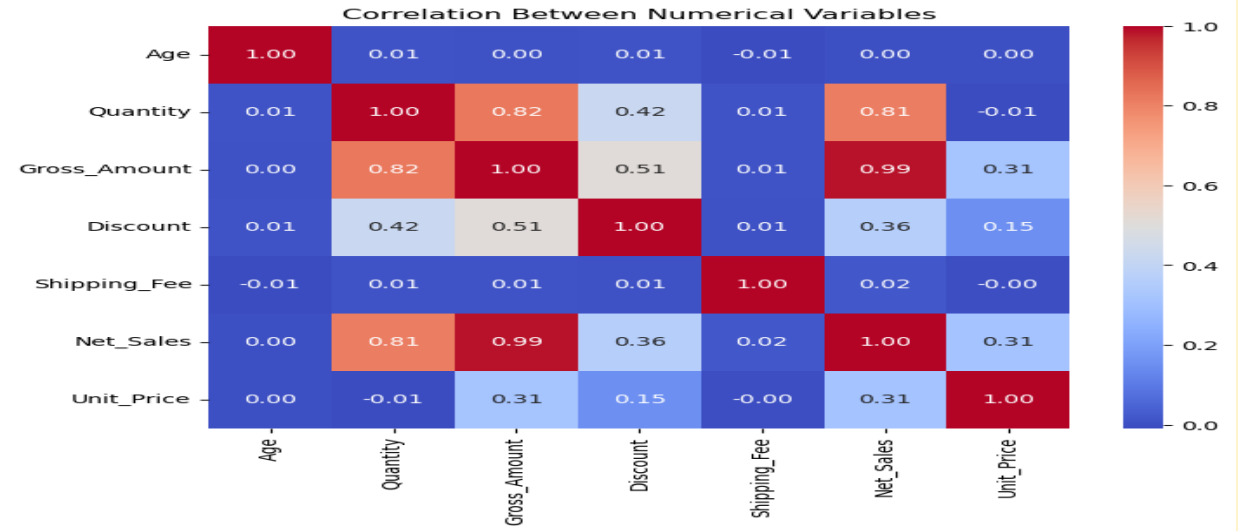


Jupyter Notebook — boxplots for Net_Sales, Quantity, and Discount

9.3 Correlation Between Numerical Variables

	Quantity	Gross_Amount	Discount	Shipping_Fee	Net_Sales
Quantity	1.00	0.82	0.42	0.01	0.81
Gross_Amount	0.82	1.00	0.51	0.01	0.99
Discount	0.42	0.51	1.00	0.01	0.36
Shipping_Fee	0.01	0.01	0.01	1.00	0.02
Net_Sales	0.81	0.99	0.36	0.02	1.00

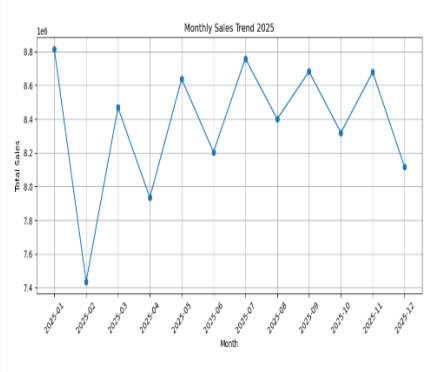
Net_Sales correlates almost perfectly with Gross_Amount (0.99, expected since it is derived from it) and strongly with Quantity (0.81). Discount shows a positive but only moderate correlation with Gross_Amount (0.51) and a weaker one with Net_Sales (0.36). This pattern is consistent with the Q7 finding: larger orders tend to carry somewhat larger discounts in absolute terms, while higher discount levels are associated with lower Net Sales per order. These correlations describe associations rather than causal relationships. Shipping_Fee is essentially uncorrelated with all other variables, consistent with a flat or fixed-fee structure. Correlation describes association, not causation — these relationships are not claimed to be causal.

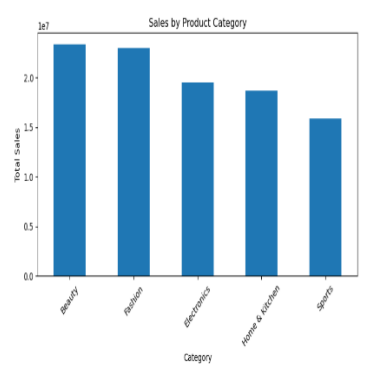


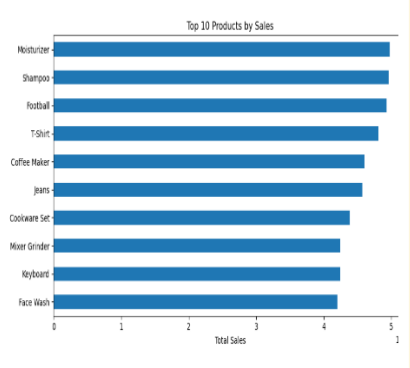
Jupyter Notebook — correlation heatmap

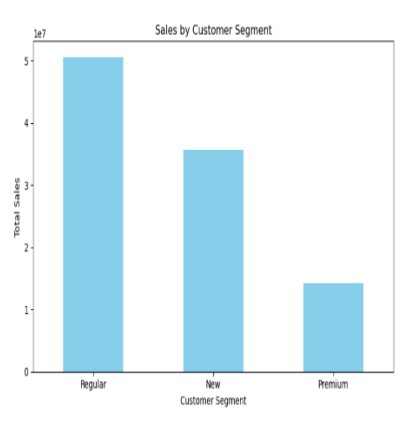
9.4 Visualizations Produced

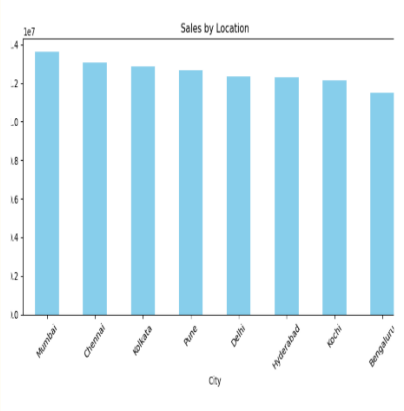
- Monthly sales trend (line chart)
- Sales by product category (bar chart)
- Top 10 products by sales (bar chart)
- Sales by customer segment (bar chart)
- Sales by location (bar chart)
- Average order value by discount band (bar chart)
- Scatter plots: Quantity vs. Net Sales, Gross Amount vs. Net Sales, Discount vs. Net Sales, Unit Price vs. Net Sales

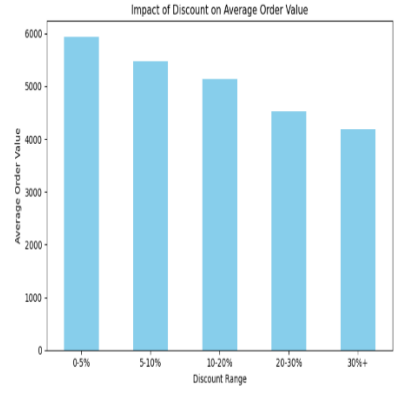


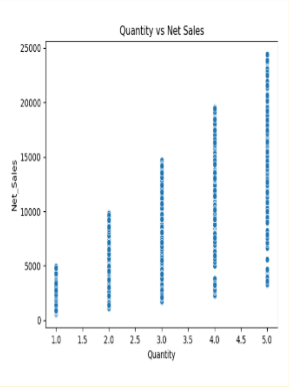


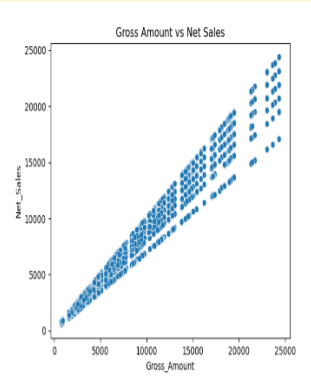


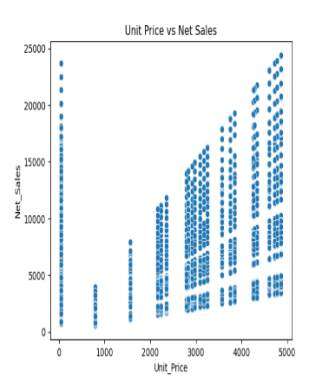


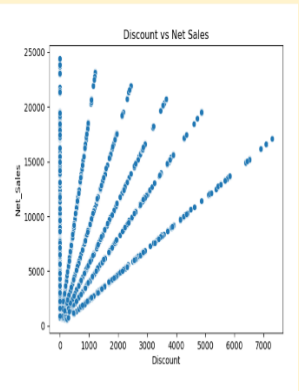












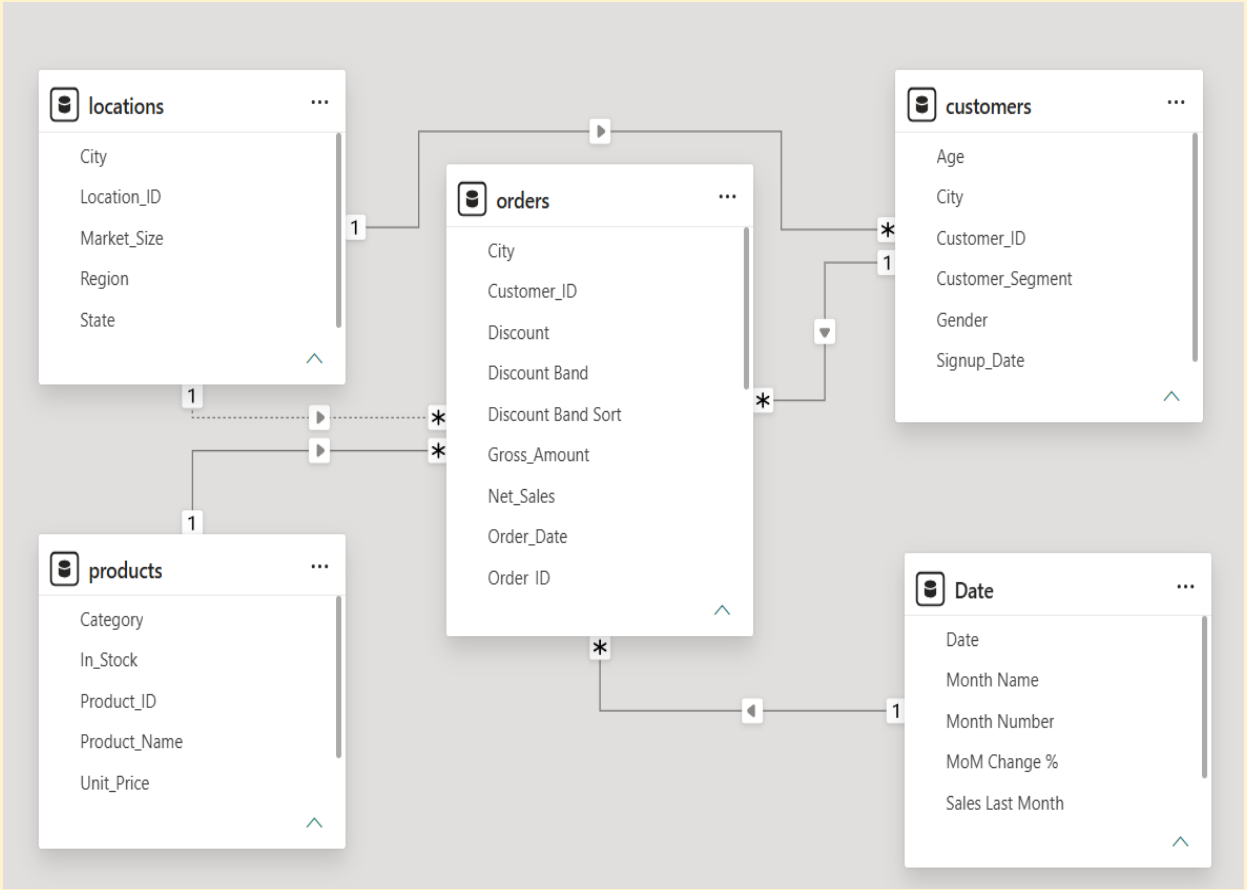
Jupyter Notebook — all six required visualizations

10. Phase E — Power BI Dashboard

An interactive four-page Power BI report was built on a star-schema data model (Orders as the fact table; Customers, Products, Locations, and a dedicated Date table as dimension tables).

10.1 Data Model

- customers[Customer_ID] → orders[Customer_ID] (one-to-many)
- products[Product_ID] → orders[Product_ID] (one-to-many)
- locations[City] → orders[City] (one-to-many)
- locations[City] → customers[City] (one-to-many)
- Date[Date] → orders[Order_Date] (one-to-many, Date marked as the official date table)



Power BI — Model view showing the star schema and relationships

10.2 DAX Measures and Calculated Columns

DAX (Data Analysis Expressions) was used in Power BI to create measures and calculated columns required for KPI calculations, time-based analysis, product ranking, and discount analysis.

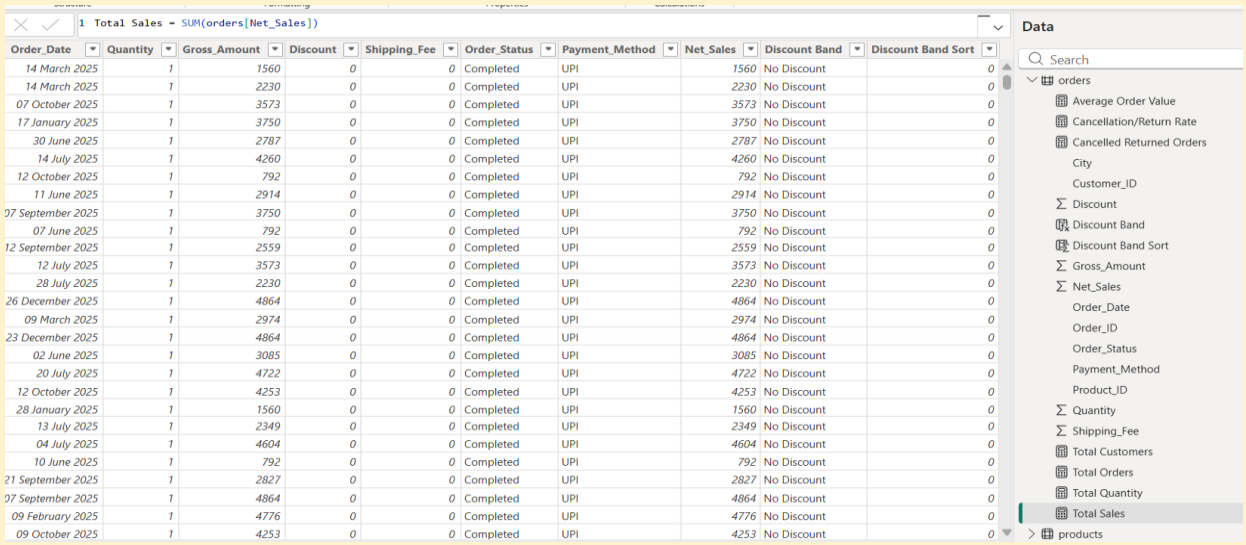
10.2.1 DAX Measures

DAX Formula	Description
`Total Sales = SUM(orders[Net_Sales])`	Calculates total Net Sales across all orders.
`Total Quantity = SUM(orders[Quantity])`	Calculates the total number of units sold.
`Total Orders = DISTINCTCOUNT(orders[Order_ID])`	Counts unique orders.
`Total Customers = DISTINCTCOUNT(orders[Customer_ID])`	Counts unique customers.
`Average Order Value = DIVIDE([Total Sales], [Total Orders])`	Calculates the average sales value per order.
`Cancelled Returned Orders = CALCULATE([Total Orders], orders[Order_Status] IN {"Cancelled", "Returned"})`	Counts orders with Cancelled or Returned status.
`Cancellation/Return Rate = DIVIDE([Cancelled Returned Orders], [Total Orders])`	Calculates the percentage of cancelled or returned orders.
`Sales Last Month = CALCULATE([Total Sales], DATEADD('Date'[Date], -1, MONTH))`	Returns total sales for the previous month.
`MoM Change % = DIVIDE([Total Sales] - [Sales Last Month], [Sales Last Month])`	Calculates month-over-month percentage change in sales.
`YTD Sales = TOTALYTD([Total Sales], 'Date'[Date])`	Calculates cumulative sales for the year to date.
`Product Rank = RANKX(ALL(products[Product_Name]), [Total Sales], , DESC, DENSE)`	Ranks products by total sales.

10.2.2 Calculated Columns and Date Table

DAX Formula	Description
`Date = CALENDAR(DATE(2025,1,1), DATE(2025,12,31))`	Creates the dedicated Date table for 2025.
`Month Name = FORMAT('Date'[Date], "MMM")`	Extracts the month name for monthly reporting.
`Month Number = MONTH('Date'[Date])`	Provides chronological month sorting.
`Discount Band = VAR pct = DIVIDE(orders[Discount], orders[Gross_Amount]) RETURN SWITCH(TRUE(), pct = 0, "No Discount", pct <= 0.10, "1-10%", pct <= 0.20, "11-20%", "21-30%")`	Groups orders into discount ranges for comparison.
`Discount Band Sort = VAR pct = DIVIDE(orders[Discount], orders[Gross_Amount]) RETURN SWITCH(TRUE(), pct = 0, 0, pct <= 0.10, 1, pct <= 0.20, 2, 3)`	Assigns numeric order to discount bands for correct sorting.

These calculations support KPI cards, monthly and year-to-date analysis, month-over-month comparison, product ranking, and discount-band analysis across the Power BI report pages.



Power BI — Fields pane showing all DAX measures, and the DAX formula view for the Total Sales measure

10.3 Report Pages

Page 1 — Executive Overview: KPI cards showing Total Sales of approximately ₹10.05 Cr, 18,000 Total Orders, Average Order Value of approximately ₹5.58K, and 4,000 Total Customers, together with the monthly sales trend.



Power BI — Page 1: Executive Overview

Page 2 — Product Performance: Top products, sales by category, quantity sold, and a product-level detail table.



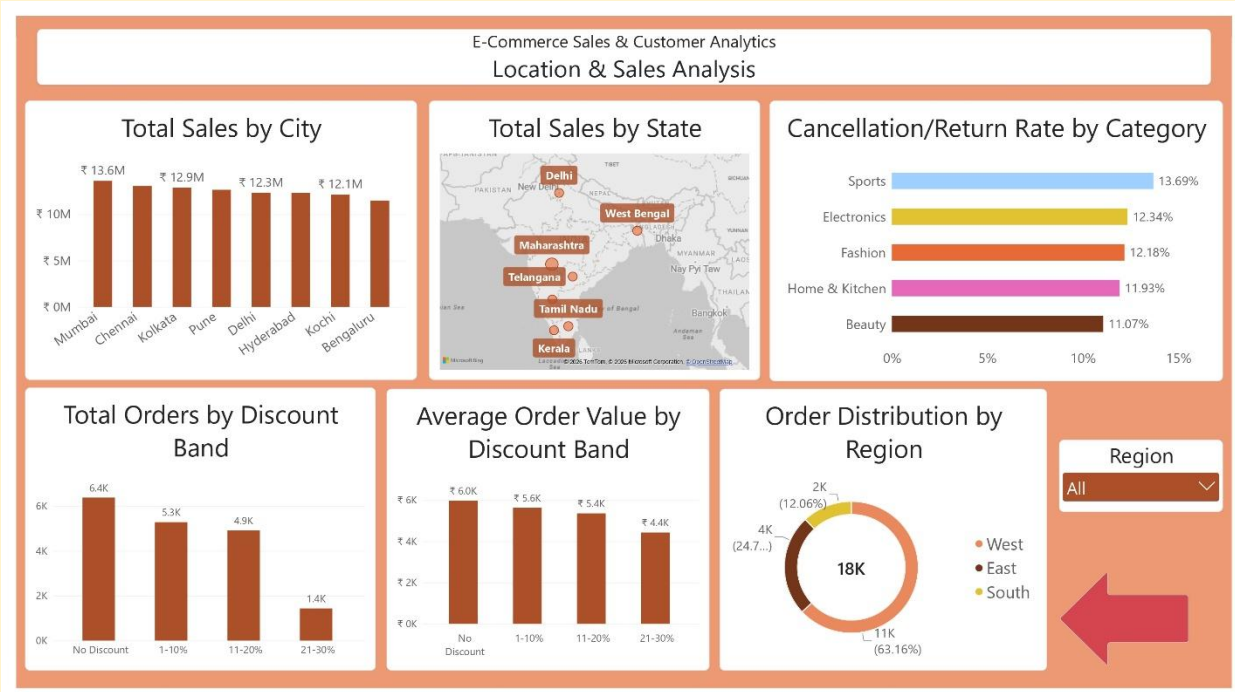
Power BI — Page 2: Product Performance

Page 3 — Customer Analysis: Sales and orders by segment, top 10 customers, and average order value by segment.



Power BI — Page 3: Customer Analysis

Page 4 — Location & Sales Analysis: Sales by city/region, order distribution, discount band analysis, and cancellation/return performance by category.



Power BI — Page 4: Location & Sales Analysis

11. Key Findings

- Data quality: 10 duplicate customers, 20 duplicate orders, 25 missing Net_Sales values, and 5 invalid product prices were identified and corrected using defensible, documented methods; two category-label case inconsistencies were also standardized.
- On an all-order basis, total sales reached ₹10,04,51,357.55 across 18,000 orders, with an Average Order Value of ₹5,580.63. January recorded the highest monthly sales and February the lowest, with no strong sustained seasonal trend.
- Beauty and Fashion lead by sales, while quantity sold is relatively evenly distributed across products, suggesting that differences in revenue are associated more with unit price than extreme demand concentration.
- Regular customers generate the highest total sales and order volume primarily because the segment is larger. Average Order Value is broadly similar across New, Regular, and Premium customers.
- Mumbai records the highest city-level sales, followed by Chennai and Kolkata, while overall sales remain relatively distributed across the eight cities.
- Average Order Value declines as discount depth increases, from approximately ₹5,972 at 0% discount to ₹4,462 at 30%, while average Gross Amount increases slightly.
- 12.24% of orders are cancelled or returned overall, with Sports showing the highest rate at 13.69% and Beauty the lowest at 11.07%.

12. Management Questions — Final Analysis

Q16. Which products and categories should management focus on?

(i)Focus on Beauty and Fashion

- **Beauty** is the strongest category with **₹2.34 crore** in sales.
- **Fashion** follows closely with **₹2.30 crore**.
- Beauty also has the highest quantity sold and number of orders.
- Management should maintain inventory availability and continue successful products/promotions in these categories.

(ii) Maintain top-performing products

- **Moisturizer:** ₹49.78 lakh — highest sales.
- **Shampoo:** ₹49.61 lakh.
- **Football:** ₹49.27 lakh.
- These products should receive priority in **inventory planning, promotions, and availability**.

(iii) Improve the Sports category

- Sports generated the **lowest category sales at ₹1.59 crore**.
- Management should identify why Sports is underperforming—such as pricing, demand, product selection, or customer preferences.
- Products such as **Cricket Bat**, which generated the lowest individual product sales (₹8.61 lakh), may need promotional or pricing strategies.

(iv) Investigate products with high orders but lower sales

- **Face Wash** has the highest quantity sold (**1,204**) and highest orders (**663**), but it is not the highest-sales product.
- This suggests that high order volume does not necessarily produce high revenue.
- Management should examine **unit price, discounts, and profitability** for such products.

(v) Control high discounts

- Your analysis shows that AOV falls from approximately **₹5,900 at 0–5% discount** to **₹4,200 at 30%+ discount**.
- Management should avoid excessive discounting unless it clearly increases sales volume or customer retention

Q17. Which customer segments generate the strongest sales?

Regular customers generate the highest total sales at approximately ₹5.06 Cr, followed by New customers at ₹3.57 Cr and Premium customers at ₹1.42 Cr. **Regular customers are the strongest revenue-generating segment**, indicating that repeat/regular customers contribute significantly to overall sales. Management should focus on **customer retention and repeat purchases** while developing strategies to convert **New customers into Regular customers**. Premium customers may require further analysis to understand their lower contribution.

Q18. Which locations have the highest business potential?

Mumbai records the highest city-level sales at approximately ₹1.36 Cr, followed by Chennai at ₹1.31 Cr and Kolkata at ₹1.29 Cr. Sales across the eight cities are relatively close, indicating a geographically diversified

customer base rather than dependence on a single city. Cities with strong sales but lower market-size classifications may warrant further investigation as potential expansion opportunities.

Focus expansion and promotional investment first on **Mumbai and Chennai**, while studying customer preferences and product performance in Bengaluru to identify opportunities for improvement.

Q19. Does discounting appear to improve sales performance?

The data does not show evidence that deeper discounts improve order value. Average Order Value declines from approximately ₹5,972 at 0% discount to ₹4,462 at 30% discount, while average Gross Amount increases only slightly. Higher discount levels are therefore associated with lower realized Net Sales per order. This is an observed relationship rather than proof of causation, so discount experiments should be evaluated using actual revenue and profitability outcomes.

Higher discounts do not appear to generate higher-value orders. Management should avoid excessive discounting and instead use targeted, moderate discounts to encourage purchases while protecting revenue. **Discounting should be controlled rather than increased**, especially at 30% or higher, unless there is evidence that it significantly increases order volume or customer retention.

Q20. What factors appear to be associated with higher sales?

Quantity and transaction value appear to be the most important practical factors associated with higher sales. Increasing the number of units purchased and promoting higher-value products may have greater impact than simply offering larger discounts.
Management should focus on **increasing order quantity, promoting high-value products, and maintaining appropriate pricing**, rather than relying heavily on discounts.

Q21. What three actions should management take based on your findings?

(i) **Focus on high-performing products and markets**

- Prioritize **Beauty and Fashion** categories and top products such as **Moisturizer and Shampoo**.
- Increase inventory and marketing efforts in strong markets, especially **Mumbai and Chennai**.

(ii) **Increase customer retention and repeat purchases**

- **Regular customers generate the highest sales (₹5.06 crore)**.
- Encourage **New customers to become Regular customers** through loyalty programs, personalized offers, and targeted campaigns.

(iii) **Control excessive discounting and improve weak performers**

- Avoid relying heavily on discounts because **30%+ discounts are associated with lower AOV**.
- Review underperforming areas such as the **Sports category and Cricket Bat**, and use targeted pricing or promotions to improve their performance.

Management should invest more in high-performing products and locations, strengthen customer retention, and use targeted rather than excessive discounting to improve overall sales and profitability.

13. Limitations & Analytical Notes

The analysis uses all 18,000 cleaned orders as the primary transaction basis, while cancellation and return performance is reported separately as an operational KPI.

Correlation analysis identifies relationships between variables but does not establish causation. In particular, the observed relationship between discount depth and Net Sales should not be interpreted as proof that discounts independently cause lower sales.

The existing customer segment labels provide useful descriptive grouping, but the similarity in Average Order Value across segments suggests that additional behavioural segmentation could provide more actionable customer insights.

The dataset covers one calendar year and eight cities, so longer-term trends, broader geographic expansion potential, customer lifetime value, and profitability effects would require additional data.

The analysis provides evidence-based directional recommendations, but operational decisions such as pricing changes, inventory allocation, and expansion should be validated through controlled testing and additional business metrics such as margin, conversion rate, and customer retention.

14. Conclusion

The analysis shows a business with stable, broad-based sales rather than one dependent on a small number of products or customers. Performance is distributed across products and cities, while customer value remains relatively similar across the existing segments.

The most actionable findings are the association between deeper discounting and lower realized order value, the elevated cancellation/return rate in Sports, and the opportunity to improve customer segmentation using behavioural measures such as recency, frequency, and monetary value.

Overall, the project demonstrates an end-to-end analytical workflow covering data cleaning, business analysis, SQL, Python, Excel, and Power BI, translating transaction-level data into practical business recommendations.